



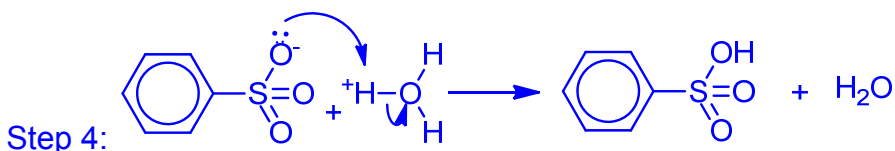
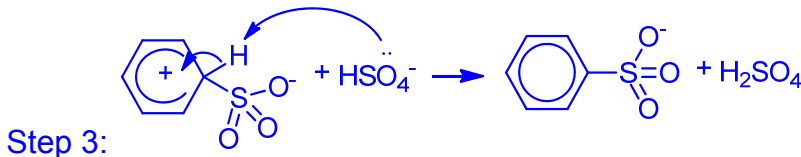
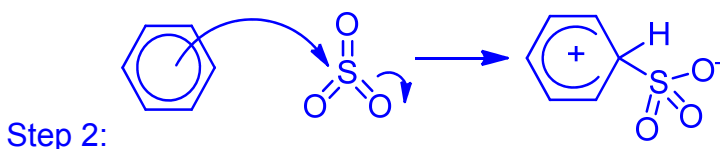
MERIDIAN JUNIOR COLLEGE
JC 2 Preliminary Examination
 H2 Chemistry Paper 3

- 1(a) (i)** Comparing experiments 1 and 2,
 When $[\text{H}_2\text{SO}_4]$ is doubled and $[\text{C}_6\text{H}_6]$ is kept constant
 Rate of reaction increased 4 times
 Order of reaction with respect to H_2SO_4 is 2

Comparing experiments 2 and 3,
 When $[\text{C}_6\text{H}_6]$ is increased 1.5 times and $[\text{H}_2\text{SO}_4]$ is kept constant
 Rate of reaction increased 1.5 times
 Order of reaction with respect to C_6H_6 is 1

(ii) $\text{Rate} = k [\text{C}_6\text{H}_6] [\text{H}_2\text{SO}_4]^2$

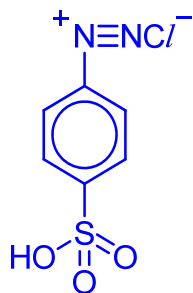
(b) (i)



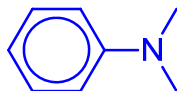
(ii) Step 2

- (c) (i)** Test: Br_2 (aq), r.t.p.
- Phenylamine: Decolourisation of orange bromine solution with the formation of a white precipitate.
 - Diazonium salt: No decolourisation of orange bromine solution.

(ii)



diazonium salt



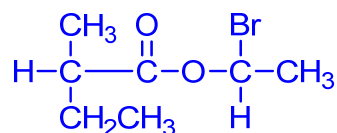
phenylamine derivative

Electrophile is the diazonium salt

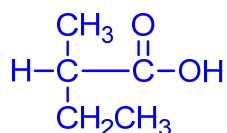
(d)

Reaction	Deduction
Q and R can rotate plane-polarized light.	Q and R contains chiral carbon and has no plane of symmetry.
0.1 mol of compound Q produces 0.5 mol of carbon dioxide gas and 0.5 mol of water.	Q contains 5 C atoms and 10 H atoms.
2 mol of Q undergoes acid-base reaction with 1 mol of Na ₂ CO ₃	Q contains 1 carboxylic acid functional group.
R undergoes oxidation when heated in an alkaline solution of iodine to give a yellow precipitate.	Yellow precipitate is CHI ₃ . R contains structure
R undergoes nucleophilic substitution when heated with sodium hydroxide, followed by the addition of dilute and aqueous silver nitrate, a cream precipitate was formed.	R contains a bromoalkane / halogenoalkane functional group. Cream precipitate is AgBr.
Q and R undergo nucleophilic acyl substitution to form P.	P contains an ester functional group

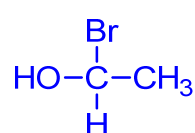
Structure of P:



Structure of Q:



Structure of R:



- 2(a) (i) Iron electrode: $\text{Mg}^{2+}(\text{l}) + 2\text{e}^- \rightarrow \text{Mg}(\text{l})$
 Graphite electrode: $2\text{Cl}^-(\text{l}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$

(ii) To prevent Mg from reaction with oxygen in the air.

(iii) Iron electrode is the cathode in which Mg is formed from molten Mg^{2+} .

As $E^\ominus_{\text{Zn}^{2+}/\text{Zn}}$ is more positive than $E^\ominus_{\text{Mg}^{2+}/\text{Mg}}$, Zn^{2+} is preferentially discharged to form Zn. Thus, the addition of ZnCl_2 will interfere with the production of Mg at the cathode.

As $E^\ominus_{\text{Ca}^{2+}/\text{Ca}}$ is less positive than that of Mg^{2+} , Ca^{2+} will not be preferentially discharged to form Mg (or Mg^{2+} will be preferentially discharged instead). Thus, the addition of CaCl_2 will not interfere with the production of Mg at the cathode.

(iv) An iron electrode is not a suitable replacement.

Fe undergoes oxidation to form $\text{Fe}^{2+}/\text{Fe}^{3+}$ instead of Cl^- oxidising to Cl_2 since $E^\ominus_{\text{Fe}^{2+}/\text{Fe}} (-0.44)$ or $E^\ominus_{\text{Fe}^{3+}/\text{Fe}} (-0.04)$ is less positive than $E^\ominus_{\text{Cl}_2/\text{Cl}^-} (+1.36)$.

$\text{Fe}^{2+}/\text{Fe}^{3+}$ formed could be reduced instead of Mg^{2+} at cathode electrode.

$$(v) \quad 1.12 \times 10^{-1} = \frac{1.5 \times 6 \times 60 \times 60}{n_e \times 96500}$$

$$n_e = 3$$

Therefore, the charge = +3

$$(b) \quad (i) \quad n_{\text{HCl}} = 0.10 \times \frac{20}{1000} = 2 \times 10^{-3} \text{ mol}$$

$$n_{\text{Mg}(\text{HCO}_3)_2} = \frac{2 \times 10^{-3}}{2} = 1 \times 10^{-3} \text{ mol}$$

$$V_{\text{Mg}(\text{HCO}_3)_2} = \frac{1 \times 10^{-3}}{0.10} = 10 \text{ cm}^3$$

$$(ii) \quad K_b = \frac{[\text{OH}^-]^2}{[\text{HCO}_3^-]}$$

$$2.4 \times 10^{-8} = \frac{[\text{OH}^-]^2}{0.10 \times 2}$$

$$[\text{OH}^-] = 6.93 \times 10^{-5} \text{ mol dm}^{-3}$$

$$\text{pH} = 14 - 4.31$$

$$\text{pH} = 9.84$$



H_2CO_3 dissociates (partially) in water or undergoes hydrolysis to produce H^+ and HCO_3^- .

(iv) Bromocresol green.

The pH transition range of the indicator (around 4 – 6) lies within the sharp pH change (around 3 – 6 from graph) over the equivalence point.

(v) 10 cm^3

$$\text{pOH} = \text{p}K_b = -\lg (2.4 \times 10^{-8})$$

$$\text{pOH} = 7.62$$

$$\text{pH} = 6.38$$

(vi) CO_3^{2-} has higher charge than HCO_3^- hence greater tendency to attract H^+ .

(c) Volume $V = 20 \text{ cm}^3$

From 0 to $V \text{ cm}^3$, the conductance decreases due to the decreasing amount of (OH^-) ions in the solution as H^+ ions react with OH^- ions to form (undissociated) water.

At $V \text{ cm}^3$, the conductance is the lowest as all the OH^- ions present have reacted with the added H^+ ions. After $V \text{ cm}^3$, as more H^+ ions are present due to the addition of hydrochloric acid, the conductance increases.

3(a) (i)

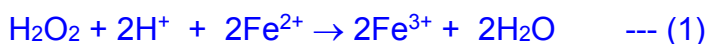


(ii) $E^{\ominus}_{\text{cell}} = +1.77 - (+0.68) = +1.09 \text{ V}$

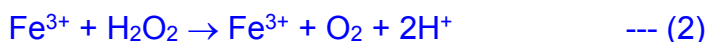
(iii) $\Delta G^{\ominus} = -(2)(96500)(1.09) = -210 \text{ kJ mol}^{-1}$

Since $\Delta G < 0$, decomposition is spontaneous.

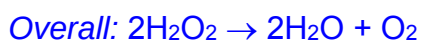
(iv) Fe^{2+} acts as a homogenous catalyst because of the ability to exist in various oxidation states, thus facilitating the formation of reaction intermediates via alternative pathways of lower E_a .



$$E^{\ominus}_{\text{cell}} = +1.77 - (+0.77) = +1.00 \text{ V} > 0$$



$$E^{\ominus}_{\text{cell}} = +0.77 - (+0.68) = +0.09 \text{ V} > 0$$



- Both steps are spontaneous since their $E^{\ominus} > 0$
- Activation energy is lower and thus reaction is faster.

(b) (i) ΔS is negative as there is a decrease in disorder due to decrease in the number of moles of gas molecules from 1 to 0.

(ii) Ag_2O

Since $\Delta G_f > 0$ for formation of Ag_2O , $\Delta G < 0$ for the decomposition of Ag_2O .



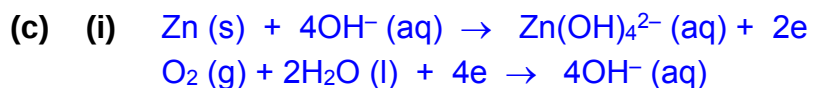
Hence, the decomposition of Ag_2O is spontaneous at 1000 K

(iii) At 2250 K,



$$\Delta G = -600 + 150 = -450 \text{ kJ mol}^{-1}$$

Since $\Delta G < 0$, the reaction is therefore thermodynamically feasible at 2250 K.



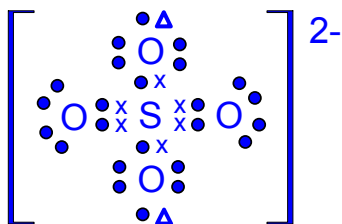
(ii) $1.59 = 0.40 - E^\ominus_{\text{oxd}}$ hence $E^\ominus_{\text{oxd}} = -1.19 \text{ V}$

(iii) $\frac{80}{100} \times \frac{1.85}{65.4} = \frac{I \times 365 \times 24 \times 60 \times 60}{2 \times 96500}$

$I = 1.38 \times 10^{-4} \text{ A}$

(iv) Zinc-air battery has a higher energy density or power over mass ratio as it uses air as an oxidising agent unlike the heavier oxidising agent MnO_2 used in the alkaline battery.

4(a) (i)



- (ii) There are 4 bond pairs and 0 lone pair around S. To minimise repulsion, the 4 electron pairs are directed towards corners of a tetrahedron hence tetrahedral shape.

(b) (i)
$$\text{Amount} = \frac{3.0}{137.3 + 2(16.0 + 1.0) + 8(18.0)} = \frac{3.0}{315.3} = 9.515 \times 10^{-3} \text{ mol}$$

Amount of H_2O lost = $9.515 \times 10^{-3} \times 9 = 8.563 \times 10^{-2} \text{ mol}$

Theoretical mass of H_2O lost = $8.563 \times 10^{-2} \times 18.0 = 1.54 \text{ g}$

Decomposition is complete.

- (ii) Ionic radius: $\text{Ba}^{2+} > \text{Mg}^{2+}$

Charge density and polarising power: $\text{Ba}^{2+} < \text{Mg}^{2+}$

Ability of cation to polarise, weaken and break O–H bond: $\text{Ba}^{2+} < \text{Mg}^{2+}$

Thermal stability: $\text{Ba}(\text{OH})_2 > \text{Mg}(\text{OH})_2$

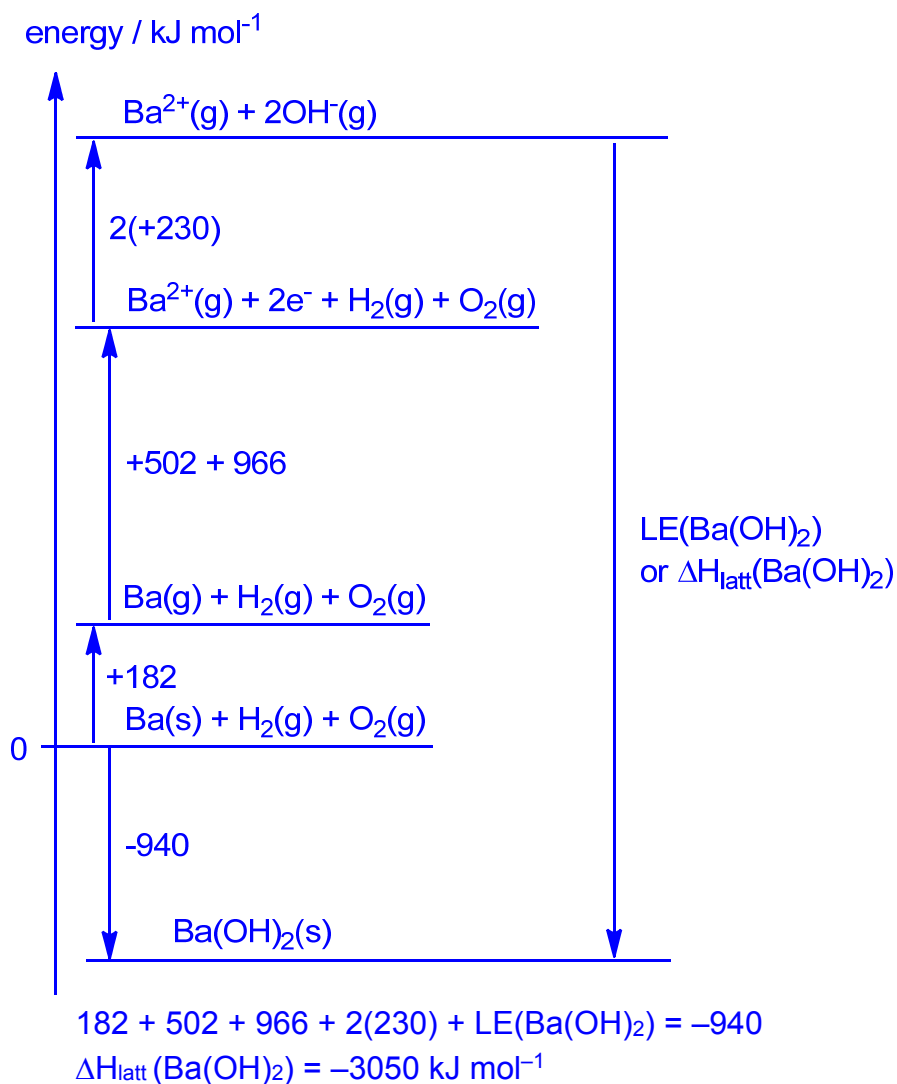
(c) (i) $34.0 - 25.0 = 9.0 \text{ }^\circ\text{C}$

(ii) Amount of H_2O formed = $\frac{100}{1000} \times 0.8 \times 2 = 0.16 \text{ mol}$

Heat absorbed by solution = $220 \times 4.18 \times 9.0 = 8276.4 \text{ J}$

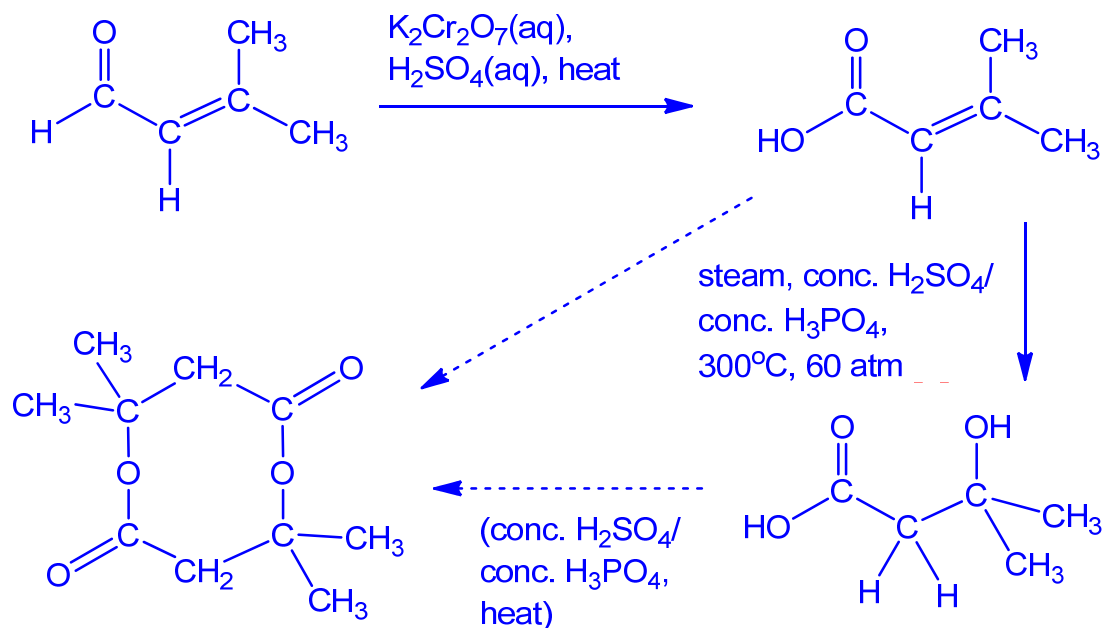
$$\Delta H_n = - \frac{8276.4}{0.16} = -51.7 \text{ kJ mol}^{-1}$$

(d)



- (e) (i) J: nucleophile
K: electrophile
- (ii) III: acid-base / Lewis base
IV: elimination / Lewis acid
- (iii) CH_3CHO and CH_3COCH_3

(iv)



5(a)



(b) (i) $\Delta H = 4 \times 410 + 3 \times 390 + \frac{3}{2} \times 496 - [410 + 890 + 6 \times 460] = -506 \text{ kJ mol}^{-1}$

ΔH (BMA process)

$= 4 \times 410 + 3 \times 390 - [410 + 890 + 3 \times 436]$

$= +202 \text{ kJ mol}^{-1}$

(ii) The Andrussov process is exothermic which is energetically more favourable

(c) $K_{sp} = s \times (2s)^2$

$$\text{Solubility} = \sqrt[3]{\frac{8.0 \times 10^{-12}}{4}}$$

$$= 1.26 \times 10^{-4} \text{ mol dm}^{-3}$$

$\text{Zn}(\text{CN})_2$ has very low solubility in water, hence the CN^- ions will not be dissociated to cause harm.

- (d) (i) Advantage: nitrite is soluble in water and hence can be transported to the body tissues quickly OR nitrite oxidises Fe^{2+} in haemoglobin to Fe^{3+} which can remove cyanide easily.

Disadvantage: the body will be starved of oxyhaemoglobin that is able to transport oxygen to the cell.

- (ii) The SCN^- is a weaker ligand that will not bind strongly to the Fe^{2+} in cytochrome c oxidase.

- (iii) Half-life = 30 min

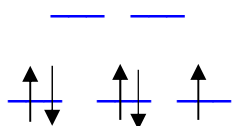
$$\text{No. of half-lives before complication occurs} = \frac{60}{30} = 2$$

$$\text{Maximum blood cyanide concentration} = 20 \times 2 \times 2 = 80 \text{ } \mu\text{g per } 100 \text{ cm}^3$$

- (iv) Two orbitals have their lobes pointing at the ligands along the x, y and z axis, hence they experienced greater repulsion from the ligands OR the ligands split the d orbitals into two energy levels.

The other three orbitals experienced less repulsion since their lobes do not point at the ligands along the axes. Hence, there are two energy levels.

(v)

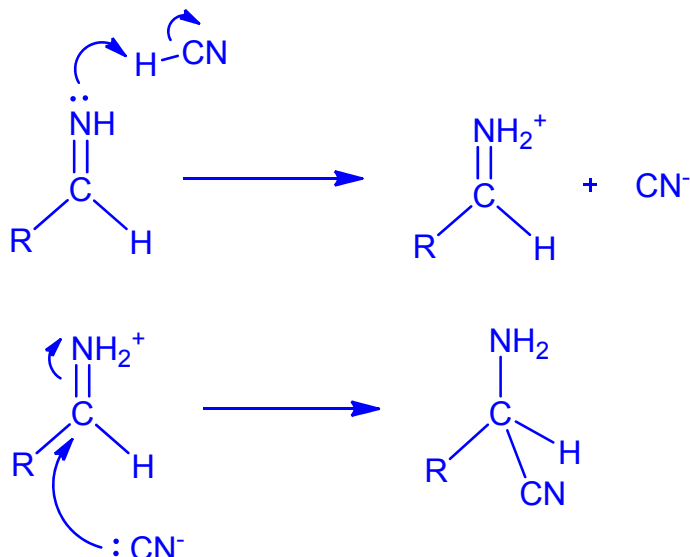


(e) (i) Elimination

(ii) NaOH (aq), heat under reflux

(iii) Prevent poisonous HCN from vapourising into the environment.

(iv)



(v) CN^- nucleophile attacks the trigonal planar carbonyl C from either side of the plane with equal probability, producing a racemic mixture. Naturally occurring amino acid is present as one of the enantiomers and will rotate plane polarised light.